

①

$$\omega_f = \omega_i + \alpha t$$

$$0 = 12 + \alpha (26)$$

$$\alpha = \frac{-12}{26} = -0.4615 \text{ RAD/s}^2$$

$$\text{MAGNITUDE: } |\alpha| = \boxed{0.46 \text{ RAD/s}^2}$$

②

THE DISK IS INITIALLY SPINNING COUNTERCLOCKWISE,
SO BY THE RIGHT-HAND RULE, ω POINTS OUT OF THE PAGE.
SINCE THE DISK IS SLOWING DOWN, α POINTS OPPOSITE TO ω .

ANSWER:

 $\boxed{\text{INTO THE PAGE}}$

③

AVERAGE ANGULAR SPEED:

$$\omega_{\text{AVG}} = \frac{\omega_i + \omega_f}{2} = \frac{(12 + 0)}{2} = 6 \text{ RAD/s}$$

ANGULAR DISPLACEMENT:

$$\Delta\theta = \omega_{\text{AVG}} t = (6)(26) = 156 \text{ RAD}$$

CONVERT TO REVOLUTIONS:

$$156 \text{ RAD} \left(\frac{1 \text{ REV}}{2\pi \text{ RAD}} \right) = 24.83 \text{ REV}$$

ANSWER:

 $\boxed{25 \text{ REVOLUTIONS}}$